



Cambridge International AS & A Level

COMPUTER SCIENCE

9618/11

Paper 1 Theory Fundamentals

May/June 2021

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **10** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)(i)	1 mark for each description Pixel: • A single square of one colour • The smallest addressable element in an image File header: • Data about the bitmap image (e.g. number of colours)	2
1(a)(ii)	1 mark per bullet point for working, 1 mark for answer Working: • $1024 \times 512 = 524\,288$ pixels/bytes • $524288 / 1024 / 1024$ Answer: 0.50 mebibytes	3
1(b)	1 mark for naming method, 1 mark per description to max 2 • Run-length encoding • Replace sequences of the same colour pixel • ... with colour code and number of identical pixels	3
1(c)(i)	252	1
1(c)(ii)	1 mark per bullet point • Converting 15 to binary 0000 1111 • Method for addition • Final answer $\begin{array}{r} 0010\ 0011 \\ +\ 0000\ 1111 \\ \hline 0011\ 0010 \\ 1\ 111 \end{array}$	3
1(c)(iii)	1 mark per bullet point • Converting -10 to two's complement binary 1111 0110 • Adding values • Final answer 0001 1001 $\begin{array}{r} 10 = 0000\ 1010 \\ -10 = 1111\ 0110 \\ \\ 0010\ 0011 \\ +\ 1111\ 0110 \\ \hline 0001\ 1001 \\ 11\ 11 \end{array}$	3

Question	Answer	Marks
1(d)	1 mark per bullet point to max 2 - The formal and legal rights to ownership // intellectual property rights - Protects against unauthorised reproduction of work - Provides for legal right of redress	2

Question	Answer	Marks
2(a)	1 mark for each correct line Utility software Disk formatter Defragmentation Back-up Disk repair Description Scans software for errors and repairs the problems Moves parts of files so that each file is contiguous in memory Creates a copy of data that is no longer required Sets up a disk so it is ready to store files Scans for errors in a disk and corrects them Creates a copy of data in case the original is lost	4
2(b)	1 mark per bullet point to max 4 - memory management - file management - security management - hardware / device / peripheral / resources management - input/output management - process management - error checking and recovery - provision of a platform for software - provision of a user interface	4

Question	Answer	Marks																																																																																																																																																																																																				
3(a)	1 mark for each completed statement The Program Counter holds the address of the next instruction to be loaded. This address is sent to the Memory Address Register. The Memory Data Register holds the data fetched from this address. This data is sent to the Current Instruction Register and the Control Unit decodes the instruction's opcode. The Program Counter is incremented.	5																																																																																																																																																																																																				
3(b)	1 mark for each shaded set of values <table><tr><th rowspan="2">Instruction address</th><th rowspan="2">ACC</th><th colspan="4">Memory address</th><th rowspan="2">IX</th><th rowspan="2">Output</th></tr><tr><th>365</th><th>366</th><th>367</th><th>368</th></tr><tr><td></td><td></td><td>1</td><td>3</td><td>65</td><td>66</td><td>0</td><td></td></tr><tr><td>200</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>201</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>202</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>203</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>204</td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>205</td><td></td><td></td><td></td><td></td><td></td><td>2</td><td></td></tr><tr><td>206</td><td>65</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>207</td><td></td><td></td><td></td><td></td><td></td><td></td><td>A</td></tr><tr><td>208</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>200</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>201</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>202</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>203</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>204</td><td></td><td>3</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>205</td><td></td><td></td><td></td><td></td><td></td><td>3</td><td></td></tr><tr><td>206</td><td>66</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>207</td><td></td><td></td><td></td><td></td><td></td><td></td><td>B</td></tr><tr><td>208</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>200</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>201</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>202</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>209</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Instruction address	ACC	Memory address				IX	Output	365	366	367	368			1	3	65	66	0		200	1							201								202								203	2							204		2						205						2		206	65							207							A	208								200	2							201								202								203	3							204		3						205						3		206	66							207							B	208								200	3							201								202								209								6
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1	1	0	1	0	1	0	0			
3(c)(ii)	1 mark for correct answer The number is divided by 8 (and only whole number retained)	1								

Question	Answer	Marks
4(a)	1 mark per bullet point to max 2 • All computers are of equal status • Each computer provides access to resources and data // data is distributed • Computers can communicate and share resources • Each computer is responsible for its own security	2
4(b)	1 mark per bullet point to max 2 per drawback • Reduced security // no central management of security • ... only as secure as the weakest computer on the network • ... each computer is at risk from viruses from other computers • No central management of backup • ... if the data from one computer is not backed up it is lost to all of them • No central management of files/software • ... consistency may be difficult to maintain • ... each computer may have different software from the others • Individual computers may respond slower • ... because they are being accessed by other computers • In order to share files etc. all the computers involved need to be switched on • ... so the files etc. may not be always available	4

Question	Answer	Marks															
4(c)(i)	1 mark for first 2 ticks, 1 mark for last 2 (shaded) <table border="1"> <thead> <tr> <th>Task</th><th>Performed by router</th><th>Not performed by router</th></tr> </thead> <tbody> <tr> <td>Receives packets from devices</td><td>✓</td><td></td></tr> <tr> <td>Finds the IP address of a Uniform Resource Locator (URL)</td><td></td><td>✓</td></tr> <tr> <td>Directs each packet to all devices attached to it</td><td></td><td>✓</td></tr> <tr> <td>Stores the IP and/or MAC address of all devices attached to it</td><td>✓</td><td></td></tr> </tbody> </table>	Task	Performed by router	Not performed by router	Receives packets from devices	✓		Finds the IP address of a Uniform Resource Locator (URL)		✓	Directs each packet to all devices attached to it		✓	Stores the IP and/or MAC address of all devices attached to it	✓		2
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4(c)(ii)	1 mark per bullet point for justification up to max 3 No mark for identification of wired/wireless Wired • Faster connection // higher bandwidth • needed as she is downloading/streaming large files • ... less time waiting / less latency / fewer delays • More reliable / stable connection • ... is less susceptible to issues with distance/walls/interference • More secure Wireless • Freedom of movement • ... can move between different rooms with a mobile device and still receive/transmit data • ... no need of a physical connection • Easily expanded if friends want to access the same network • Less cabling / expertise is needed • ... making the initial setup less expensive	3															
4(d)	1 mark for identifying that she is using both. 1 mark per bullet point for justification • using internet because sending data on the infrastructure • using WWW because accessing a website (that is stored on a web server operated by the webmail) that is part of the WWW	3															

Question	Answer	Marks
5(a)	1 mark per bullet point to max 2 - Definition: Microprocessor/microcontroller within a larger system // microprocessor/microcontroller that performs one specific task - Example: e.g. Embedded system in washing machine only controls the programs for the washing cycle // it is part of the washing machine but does not perform any other function within it	2
5(b)	1 mark for RAM, 1 mark for ROM RAM: Store the choices/wash program the user has entered // stores the data read from the sensors // stores the time left in the program // by example ROM: Store the start-up instructions (for the washing cycles)	2
5(c)	1 mark per bullet point - The system uses feedback - The system causes the temperature to change // produces an action	2

Question	Answer	Marks
6(a)	Range (check)	1
6(b)	Presence (check)	1
6(c)	Existence (check)	1

Question	Answer	Marks								
7(a)	1 mark per bullet point to max 3 - Flat-file has more data redundancy ... because the same data is stored many times // data is stored in different tables which are linked - There is program-data dependence with flat-files ... because any changes to the structure of the data means the programs that access that data have to be re-written - Flat-file has more data inconsistency // worse data integrity ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere - It is not easy to perform complex searches /queries ... because a new program has to be written each time - Flat files could have a lack of privacy ... as user views cannot easily be implemented	3								
7(b)(i)	1 mark for each correct example one-to-one - e.g. customer to payment details // customer to login details one-to-many - e.g. customer to order many-to-many - e.g. order to product // customer to product	3								
7(b)(ii)	1 mark <table border="1"><thead><tr><th>Relationship</th><th>Tick (✓)</th></tr></thead><tbody><tr><td>one-to-one</td><td></td></tr><tr><td>one-to-many</td><td></td></tr><tr><td>many-to-many</td><td>✓</td></tr></tbody></table>	Relationship	Tick (✓)	one-to-one		one-to-many		many-to-many	✓	1
Relationship	Tick (✓)									
one-to-one										
one-to-many										
many-to-many	✓									
7(b)(iii)	1 mark CREATE DATABASE SHOPORDERS ;	1								
7(c)	1 mark per item to max 3 - table name - field name // attribute - data type - type of validation - Primary Key - Foreign Key - relationships	3								

Question	Answer						Marks
8	1 mark per correct row						3
	Statement	AND	NAND	NOR	XOR	OR	
	The output is 1 only when both inputs are 1	✓					
	The output is 1 only when both inputs are different				✓		
	The output is 1 only when both inputs are 0			✓			



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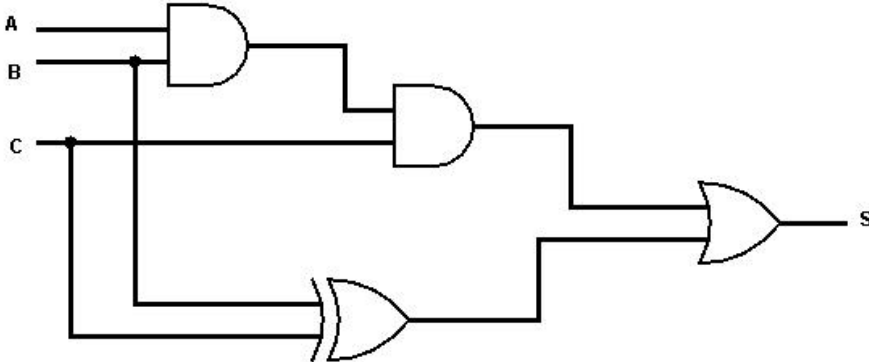
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Question	Answer	Marks								
1(a)	1 mark for definition, 1 mark for appropriate example in each <table><tr><td>Term</td><td>Definition and example</td></tr><tr><td>Field</td><td>A column/attribute in a table e.g. CustomerID in the table CUSTOMER</td></tr><tr><td>Entity</td><td>Anything that data can be stored about e.g. A customer or a house</td></tr><tr><td>Foreign Key</td><td>A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u></td></tr></table>	Term	Definition and example	Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER	Entity	Anything that data can be stored about e.g. A customer or a house	Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>	6
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1(b)	1 mark per bullet point to max 2 - All fields in all tables are dependant fully on the PK and on no other fields - for example all fields in Customer table are fully dependent on CustomerID	2								
1(c)(i)	1 mark for each correctly completed line <pre>CREATE TABLE RENTAL (RentalID INTEGER NOT NULL, CustomerID INTEGER NOT NULL, HouseID VARCHAR (5) NOT NULL, MonthlyCost REAL/CURRENCY NOT NULL, DepositPaid BOOLEAN NOT NULL, PRIMARY KEY (RentalID));</pre>	4								
1(c)(ii)	1 mark per bullet point - Select FirstName and LastName - From both tables - Where DepositPaid = No - Joining tables (either AND, or INNER JOIN) Example script: <pre>SELECT FirstName, LastName FROM CUSTOMER, RENTAL WHERE DepositPaid = No AND RENTAL.CustomerID = CUSTOMER.CustomerID;</pre>	4								

Question	Answer	Marks																									
2(a)	1 mark per bullet point to max 2 • To make sure the team members feel valued • To get the best work out of the team • To enable them to work well together • To enable them to create the best product for the client	2																									
2(b)	1 mark per bullet point to max 3 • The rules / past moves / decision making algorithms of the game will be stored • The AI program is trained, by playing many times • AI will look (ahead) at possible moves • ... and/or analyse the pattern of past choices • ... and choose the move most likely to be successful • Computer could learn how to improve // learn from previous mistakes • ... by storing the positive/negative result of choices • ... and changing its future choices	3																									
2(c)	1 mark for each correct column <table><tr><th>Statement</th><th>Free Software Foundation</th><th>Open Source Initiative</th><th>Shareware</th><th>Commercial Software</th></tr><tr><td>The user can edit the source code</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>The user must always pay before being able to use the software</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>The user can redistribute the software</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>The user always gets a trial period</td><td></td><td></td><td>✓</td><td></td></tr></table>	Statement	Free Software Foundation	Open Source Initiative	Shareware	Commercial Software	The user can edit the source code	✓	✓			The user must always pay before being able to use the software				✓	The user can redistribute the software	✓	✓	✓		The user always gets a trial period			✓		4
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3(a)	1 mark for each correct gate, with correct inputs 	4																																													
3(b)	1 mark for each half (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>S</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	S	0	0	0		0	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		1	2
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4(a)	1 mark for each shaded section / bullet point • Load 65 into ACC • Load 100 into ACC, increment and store in 102 • Load 68 into ACC • Load 101 into ACC, decrement and store in 102 <table><tr><th rowspan="2">Instruction address</th><th rowspan="2">ACC</th><th colspan="3">Memory address</th></tr><tr><th>100</th><th>101</th><th>102</th></tr><tr><td></td><td></td><td>68</td><td>65</td><td>100</td></tr><tr><td>70</td><td>65</td><td></td><td></td><td></td></tr><tr><td>71</td><td></td><td></td><td></td><td></td></tr><tr><td>72</td><td></td><td></td><td></td><td></td></tr><tr><td>73</td><td></td><td></td><td></td><td></td></tr><tr><td>74</td><td></td><td></td><td></td><td></td></tr><tr><td>76</td><td>100</td><td></td><td></td><td></td></tr><tr><td>77</td><td>101</td><td></td><td></td><td></td></tr><tr><td>78</td><td></td><td></td><td></td><td>101</td></tr><tr><td>79</td><td></td><td></td><td></td><td></td></tr><tr><td>70</td><td>68</td><td></td><td></td><td></td></tr><tr><td>71</td><td></td><td></td><td></td><td></td></tr><tr><td>72</td><td></td><td></td><td></td><td></td></tr><tr><td>80</td><td>101</td><td></td><td></td><td></td></tr><tr><td>81</td><td>100</td><td></td><td></td><td></td></tr><tr><td>82</td><td></td><td></td><td></td><td>100</td></tr><tr><td>83</td><td></td><td></td><td></td><td></td></tr><tr><td>(70)</td><td></td><td></td><td></td><td></td></tr></table>	Instruction address	ACC	Memory address			100	101	102			68	65	100	70	65				71					72					73					74					76	100				77	101				78				101	79					70	68				71					72					80	101				81	100				82				100	83					(70)					4
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Question	Answer	Marks
4(b)(iii)	1 mark for AND, 1 mark for #15 AND #15	2

Question	Answer	Marks
5(a)	1 mark for each term correctly inserted The control unit/bus transmits the signals to coordinate events based on the pulses of the (system) clock . The data bus carries data to components, while the address bus carries the address where data is being written to or read from. The arithmetic logic unit/ALU performs mathematical operations and logical comparisons.	5
5(b)	1 mark per bullet point to max 3 per factor. max 4 overall. Number of cores: - Each core processes one <u>instruction</u> per clock pulse - More/multiple cores mean that sequences of instructions can be split between them ... and so more than one <u>instruction</u> is executed per clock pulse // more sequences of instructions can be run at the same time More cores decreases the time taken to complete task Clock speed: - Each <u>instruction</u> is executed on a clock pulse // one F-E cycle is run on each clock pulse ... so the clock speed dictates the number of <u>instructions</u> that can be run per second - The faster the clock speed the more <u>instructions</u> can be run per second	4
5(c)(i)	1 mark per bullet point to max 2 - Cloud storage can be free (for small quantities) - No need for separate (high capacity) storage devices // saves storage on existing devices - Can access data from any computer with internet access - Most cloud data services will have in-built backup/disaster recovery - Security could be better - Can easily increase capacity - Data can be easily shared	2

Question	Answer	Marks														
5(c)(ii)	1 mark per bullet point to max 2: • Can only access (the cloud) with internet access • Security may not be strong // no control over security • There may not be any backups // no control over backups • It can take a long time to upload/download the data • It can be more expensive in the long term • There could be a limit to the amount of storage unless paid for • There could be compatibility/access issues • There could be issues with the company offering cloud services	2														
5(d)	1 mark for each correct line <table><thead><tr><th>Term</th><th>Description</th></tr></thead><tbody><tr><td>Public IP Address</td><td>It is only visible to devices within the Local Area Network (LAN)</td></tr><tr><td>Private IP address</td><td>It increments by 1 each time the device connects to the internet</td></tr><tr><td>Dynamic IP address</td><td>A new one is reallocated each time a device connects to the internet</td></tr><tr><td>Static IP address</td><td>It can only be allocated to a router</td></tr><tr><td></td><td>It is visible to any device on the internet</td></tr><tr><td></td><td>It does not change each time a device is connected to the internet</td></tr></tbody></table>	Term	Description	Public IP Address	It is only visible to devices within the Local Area Network (LAN)	Private IP address	It increments by 1 each time the device connects to the internet	Dynamic IP address	A new one is reallocated each time a device connects to the internet	Static IP address	It can only be allocated to a router		It is visible to any device on the internet		It does not change each time a device is connected to the internet	4
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Question	Answer	Marks
6(a)	1 mark for each correct answer ASCII = 128 // 2^7 Extended ASCII = 256 // 2^8	2
6(b)	1 mark per bullet point to max 2 - Each character has its own unique code - Each character in the word is replaced by its code - The codes are stored in the order in the word	2

Question	Answer	Marks
6(c)(i)	31	1
6(c)(ii)	53	1

Question	Answer	Marks
7(a)	1 mark per bullet point to max 2 - Program libraries store pre-written functions and routines - The program library can be referenced/imported - the functions/routines can be called in her own program	2
7(b)(i)	1 mark per bullet point to max 4; max 3 from each section Interpreter: - Use an interpreter while writing the program ... to test/debug the partially completed program ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: - Use the compiler after the program is complete ... to create an executable file - Use the compiler to repeatedly test the same (completed) section ... without having to re-interpret every time // compiler not needed at run-time	4
7(b)(ii)	1 mark per correct tool to max 2 e.g. - Breakpoints - Single stepping - Report windows	2

Question	Answer	Marks
8(a)	1 mark per bullet point - Security protects data against loss - Privacy protects data against unauthorised access	2
8(b)	1 mark for a correct answer - Two factor authentication - Biometric passwords - Key Card Access - Firewall	1

Question	Answer	Marks
8(c)	1 mark per correct answer to max 2 • Malware // viruses // spyware // by example • Hacking • Phishing • Pharming	2